300 INTERVIEW QUESTIONS

Foundation Data Engineering | Medium + Advanced

Six modules. 50 questions per module. Concise subject-matter-expert answers with specific fresher interview expectations.

01 Programming Logic and Fundamentals | 50 questions

02 SQL and Database Management | 50 questions

03 Data Warehousing, Snowflake and Data Modeling | 50 questions

04 Python Programming and Software Design | 50 questions

05 Python Analytics, Integration and Projects | 50 questions

06 Apache Spark and Distributed Processing | 50 questions

How to use: First answer aloud in 30-60 seconds; then compare with the model answer. For advanced questions, expand your response with a concrete example, assumptions and a validation method.

Difficulty: In each module, Q1-Q25 are medium and Q26-Q50 are advanced application or troubleshooting prompts. Advanced describes question depth, not expected years of experience.

MODULE 01 | Programming Logic and Fundamentals

What interviewers expect: correct terminology, a brief explanation, a practical illustration and awareness of common pitfalls.

MEDIUM | Concept clarity and core usage (Q1-Q25)

01. How does pseudocode help before coding?

SME answer: It expresses a solution as language-independent steps and exposes missing decisions early.

Interviewer expects: *Explain the definition accurately and give a simple business example.*

02. What is the difference between a flowchart and pseudocode?

SME answer: A flowchart depicts control flow visually; pseudocode describes the same logic in structured text.

Interviewer expects: *Contrast the behavior with a related concept and identify the key limitation.*

03. How do you validate an algorithm?

SME answer: Dry-run representative inputs, edge cases and expected outputs before implementation.

Interviewer expects: *Describe a small working example and its expected output.*

04. What is a variable?

SME answer: A named reference or storage location for a value whose state can change.

Interviewer expects: *State when the concept is useful and one common mistake.*

05. What is the difference between an integer and a floating-point number?

SME answer: Integers represent whole numbers; floating-point values approximate real numbers within finite precision.

Interviewer expects: *Connect the concept to correctness, maintainability or performance.*

06. What is type conversion?

SME answer: Converting data between types, explicitly or implicitly depending on the language.

Interviewer expects: *Explain the definition accurately and give a simple business example.*

07. When would you use a Boolean?

SME answer: To represent a true/false condition such as eligibility or validation status.

Interviewer expects: *Contrast the behavior with a related concept and identify the key limitation.*

08. What is operator precedence?

SME answer: Rules determining the order in which operators are evaluated when parentheses are absent.

Interviewer expects: *Describe a small working example and its expected output.*

09. How do AND and OR differ?

SME answer: AND requires all conditions to be true; OR requires at least one condition to be true.

Interviewer expects: *State when the concept is useful and one common mistake.*

10. When should you use if-elif-else?

SME answer: When mutually exclusive conditions need different execution paths.

Interviewer expects: *Connect the concept to correctness, maintainability or performance.*

11. How does a for loop differ from a while loop?

SME answer: A for loop typically iterates over a sequence; a while loop repeats while a condition holds.

Interviewer expects: *Explain the definition accurately and give a simple business example.*

12. What is an infinite loop?

SME answer: A loop whose terminating condition is never reached unless externally interrupted.

Interviewer expects: *Contrast the behavior with a related concept and identify the key limitation.*

13. Why are loop counters important?

SME answer: They track progress and help prevent off-by-one mistakes and nontermination.

Interviewer expects: *Describe a small working example and its expected output.*

14. What does break do?

SME answer: It exits the nearest enclosing loop immediately.

Interviewer expects: *State when the concept is useful and one common mistake.*

15. What does continue do?

SME answer: It skips the remaining body of the current loop iteration.

Interviewer expects: *Connect the concept to correctness, maintainability or performance.*

16. What is a function?

SME answer: A reusable block of logic with an interface of parameters and optional return values.

Interviewer expects: *Explain the definition accurately and give a simple business example.*

17. What is the difference between a parameter and an argument?

SME answer: A parameter is declared by a function; an argument is passed during a call.

Interviewer expects: *Contrast the behavior with a related concept and identify the key limitation.*

18. Why return a value instead of printing it?

SME answer: Returned values can be tested, reused and composed in later calculations.

Interviewer expects: *Describe a small working example and its expected output.*

19. What is modular programming?

SME answer: Decomposing a system into small, independently understandable and reusable units.

Interviewer expects: *State when the concept is useful and one common mistake.*

20. What is an algorithm?

SME answer: A finite, precise sequence of steps that transforms input into output.

Interviewer expects: *Connect the concept to correctness, maintainability or performance.*

21. What is an edge case?

SME answer: An unusual boundary input that can expose incorrect assumptions.

Interviewer expects: *Explain the definition accurately and give a simple business example.*

22. What does debugging mean?

SME answer: Systematically locating the cause of incorrect behavior and verifying a fix.

Interviewer expects: *Contrast the behavior with a related concept and identify the key limitation.*

23. What is the difference between syntax and logic errors?

SME answer: Syntax errors violate language grammar; logic errors execute but produce incorrect results.

Interviewer expects: *Describe a small working example and its expected output.*

24. What is an array index?

SME answer: A position used to access a sequence element; index conventions differ by language.

Interviewer expects: *State when the concept is useful and one common mistake.*

25. What is a trace table?

SME answer: A table recording variable states at each important execution step.

Interviewer expects: *Connect the concept to correctness, maintainability or performance.*

ADVANCED | Scenarios, tradeoffs and troubleshooting (Q26-Q50)

26. How do you find the second-largest distinct number in a list?

SME answer: Track the largest and second-largest distinct values in one pass, handling missing results.

Interviewer expects: *Describe a realistic failure mode and the steps to diagnose it.*

27. How do you detect duplicates efficiently?

SME answer: Use a hash set while traversing elements; expected average lookup is constant time.

Interviewer expects: *Justify the choice against alternatives and state tradeoffs.*

28. How would you reverse a string without a built-in reverse function?

SME answer: Walk indices from the end to the beginning and append characters to an output buffer.

Interviewer expects: *Specify assumptions, boundary conditions and validation checks.*

29. How do you verify a palindrome?

SME answer: Compare characters from both ends until pointers cross, after any required normalization.

Interviewer expects: *Explain the behavior on realistic datasets and edge cases.*

30. How would you count word frequencies?

SME answer: Normalize/tokenize text and increment each token's count in a dictionary.

Interviewer expects: *Discuss performance, concurrency or operational consequences when relevant.*

31. How do you handle division by zero in a calculator?

SME answer: Validate the denominator and return a clear error rather than attempting division.

Interviewer expects: *Describe a realistic failure mode and the steps to diagnose it.*

32. How do you design a login-attempt limit?

SME answer: Maintain an attempt counter, validate credentials and stop at a defined threshold.

Interviewer expects: *Justify the choice against alternatives and state tradeoffs.*

33. What is an off-by-one error?

SME answer: An incorrect boundary condition that executes one iteration too many or too few.

Interviewer expects: *Specify assumptions, boundary conditions and validation checks.*

34. How can you avoid nested conditional complexity?

SME answer: Use guard clauses, helper functions and clear business rules.

Interviewer expects: *Explain the behavior on realistic datasets and edge cases.*

35. What is time complexity?

SME answer: The asymptotic growth of an algorithm's computational work as input size increases.

Interviewer expects: *Discuss performance, concurrency or operational consequences when relevant.*

36. What is O(n) versus O(n squared)?

SME answer: O(n) grows linearly with input size; O(n squared) grows quadratically.

Interviewer expects: *Describe a realistic failure mode and the steps to diagnose it.*

37. What is space complexity?

SME answer: The memory growth of an algorithm relative to its input size.

Interviewer expects: *Justify the choice against alternatives and state tradeoffs.*

38. How does binary search work?

SME answer: It halves a sorted search interval until the target is found or the interval is empty.

Interviewer expects: *Specify assumptions, boundary conditions and validation checks.*

39. When is binary search inappropriate?

SME answer: When the search domain is not ordered or random access is too costly.

Interviewer expects: *Explain the behavior on realistic datasets and edge cases.*

40. How would you find the missing number from 1 to n?

SME answer: Compare the expected arithmetic sum with the observed sum when uniqueness assumptions hold.

Interviewer expects: *Discuss performance, concurrency or operational consequences when relevant.*

41. How do you handle null or missing inputs?

SME answer: Define behavior explicitly: reject, default, propagate or skip according to the contract.

Interviewer expects: *Describe a realistic failure mode and the steps to diagnose it.*

42. How do you validate a date input?

SME answer: Check parsing, calendar validity and any business range constraints.

Interviewer expects: *Justify the choice against alternatives and state tradeoffs.*

43. What is recursion?

SME answer: A function solving a problem by calling itself on smaller instances until a base case.

Interviewer expects: *Specify assumptions, boundary conditions and validation checks.*

44. Why is a recursion base case essential?

SME answer: It terminates self-calls and prevents unbounded recursion.

Interviewer expects: *Explain the behavior on realistic datasets and edge cases.*

45. When would you choose iteration over recursion?

SME answer: Use iteration for simple repeated state changes or when stack depth is a concern.

Interviewer expects: *Discuss performance, concurrency or operational consequences when relevant.*

46. What is defensive programming?

SME answer: Validating inputs, stating invariants and managing anticipated failures deliberately.

Interviewer expects: *Describe a realistic failure mode and the steps to diagnose it.*

47. How do you make code testable?

SME answer: Separate pure logic from I/O, use small functions and predictable outputs.

Interviewer expects: *Justify the choice against alternatives and state tradeoffs.*

48. What is a pure function?

SME answer: A function with no observable side effects whose output depends only on inputs.

Interviewer expects: *Specify assumptions, boundary conditions and validation checks.*

49. How do you explain a solution in an interview?

SME answer: State assumptions, outline logic, analyze complexity and demonstrate boundary tests.

Interviewer expects: *Explain the behavior on realistic datasets and edge cases.*

50. How do you evaluate tradeoffs between two algorithms?

SME answer: Compare correctness, asymptotic complexity, memory, readability and actual input sizes.

Interviewer expects: *Discuss performance, concurrency or operational consequences when relevant.*

MODULE 02 | SQL and Database Management

What interviewers expect: correct terminology, a brief explanation, a practical illustration and awareness of common pitfalls.

MEDIUM | Concept clarity and core usage (Q1-Q25)

01. What is a primary key?

SME answer: A non-null unique identifier for each table row; it may comprise multiple columns.

Interviewer expects: *Explain the definition accurately and give a simple business example.*

02. What is a foreign key?

SME answer: A referential constraint requiring non-null referencing values to match an eligible parent key.

Interviewer expects: *Contrast the behavior with a related concept and identify the key limitation.*

03. What is the difference between INNER JOIN and LEFT JOIN?

SME answer: INNER JOIN returns matching pairs; LEFT JOIN retains unmatched left rows with NULL right columns.

Interviewer expects: *Describe a small working example and its expected output.*

04. What does FULL OUTER JOIN return?

SME answer: All matched rows plus unmatched rows from both inputs, padded with NULLs.

Interviewer expects: *State when the concept is useful and one common mistake.*

05. What is a SELF JOIN?

SME answer: A table joined to itself using aliases to model relationships such as employee-manager.

Interviewer expects: *Connect the concept to correctness, maintainability or performance.*

06. What is WHERE versus HAVING?

SME answer: WHERE filters input rows before grouping; HAVING filters grouped results.

Interviewer expects: *Explain the definition accurately and give a simple business example.*

07. How does GROUP BY work?

SME answer: It groups rows by expressions so aggregate functions can compute per-group results.

Interviewer expects: *Contrast the behavior with a related concept and identify the key limitation.*

08. What is COUNT(*) versus COUNT(column)?

SME answer: COUNT(*) counts rows; COUNT(column) excludes rows where that column is NULL.

Interviewer expects: *Describe a small working example and its expected output.*

09. What is a subquery?

SME answer: A SELECT expression nested within another statement, sometimes correlated to outer rows.

Interviewer expects: *State when the concept is useful and one common mistake.*

10. What is a CTE?

SME answer: A named query expression scoped to one statement, introduced with WITH.

Interviewer expects: *Connect the concept to correctness, maintainability or performance.*

11. What is UNION versus UNION ALL?

SME answer: UNION removes duplicate output rows; UNION ALL retains duplicates.

Interviewer expects: *Explain the definition accurately and give a simple business example.*

12. What does COALESCE do?

SME answer: It returns the first non-NULL argument, using database-specific type resolution.

Interviewer expects: *Contrast the behavior with a related concept and identify the key limitation.*

13. What is the difference between DELETE and TRUNCATE?

SME answer: DELETE can filter rows; TRUNCATE removes all rows via a DB-specific bulk operation with varying transactional behavior.

Interviewer expects: *Describe a small working example and its expected output.*

14. What is an index?

SME answer: A structure that can speed retrieval but adds storage and write-maintenance overhead.

Interviewer expects: *State when the concept is useful and one common mistake.*

15. What is a view?

SME answer: A stored query definition presenting a logical table; materialization depends on view type.

Interviewer expects: *Connect the concept to correctness, maintainability or performance.*

16. What are ACID properties?

SME answer: Atomicity, Consistency, Isolation and Durability describe transaction guarantees.

Interviewer expects: *Explain the definition accurately and give a simple business example.*

17. What does COMMIT do?

SME answer: It makes changes in the current transaction durable under the database's guarantees.

Interviewer expects: *Contrast the behavior with a related concept and identify the key limitation.*

18. What does ROLLBACK do?

SME answer: It reverses uncommitted changes in a transaction, subject to DBMS-specific rules.

Interviewer expects: *Describe a small working example and its expected output.*

19. What is a SAVEPOINT?

SME answer: A named point allowing partial rollback within an active transaction.

Interviewer expects: *State when the concept is useful and one common mistake.*

20. What is GRANT versus REVOKE?

SME answer: GRANT assigns privileges; REVOKE removes specified privileges according to DBMS rules.

Interviewer expects: *Connect the concept to correctness, maintainability or performance.*

21. What is normalization?

SME answer: Structuring relational data to reduce redundancy and update anomalies.

Interviewer expects: *Explain the definition accurately and give a simple business example.*

22. What is denormalization?

SME answer: Intentionally introducing redundancy or precomputation to improve selected read patterns.

Interviewer expects: *Contrast the behavior with a related concept and identify the key limitation.*

23. What is a window function?

SME answer: An analytic calculation over related rows that preserves row-level output.

Interviewer expects: *Describe a small working example and its expected output.*

24. What is ROW_NUMBER versus RANK?

SME answer: ROW_NUMBER uniquely sequences rows; RANK gives ties equal ranks and leaves gaps.

Interviewer expects: *State when the concept is useful and one common mistake.*

25. What is the difference between RANK and DENSE_RANK?

SME answer: Both preserve ties; DENSE_RANK does not skip subsequent rank values.

Interviewer expects: *Connect the concept to correctness, maintainability or performance.*

ADVANCED | Scenarios, tradeoffs and troubleshooting (Q26-Q50)

26. How do you return the second-highest distinct salary?

SME answer: Use DENSE_RANK over salary descending and filter rank 2, or select MAX below MAX.

Interviewer expects: *Describe a realistic failure mode and the steps to diagnose it.*

27. How do you find duplicate business keys?

SME answer: GROUP BY the business key and filter HAVING COUNT(*) > 1.

Interviewer expects: *Justify the choice against alternatives and state tradeoffs.*

28. How do you remove duplicate rows but retain the newest?

SME answer: Rank rows by business key and timestamp with ROW_NUMBER, then delete ranks greater than one.

Interviewer expects: *Specify assumptions, boundary conditions and validation checks.*

29. How do you calculate a running total?

SME answer: Use SUM(amount) OVER (PARTITION BY key ORDER BY date ROWS UNBOUNDED PRECEDING).

Interviewer expects: *Explain the behavior on realistic datasets and edge cases.*

30. How do you get top three salaries per department?

SME answer: Apply DENSE_RANK or ROW_NUMBER partitioned by department, then filter ranks at most three.

Interviewer expects: *Discuss performance, concurrency or operational consequences when relevant.*

31. What is the difference between NOT IN and NOT EXISTS with NULL?

SME answer: NOT IN can evaluate UNKNOWN when the subquery contains NULL; NOT EXISTS avoids that trap.

Interviewer expects: *Describe a realistic failure mode and the steps to diagnose it.*

32. How can a LEFT JOIN become an INNER JOIN accidentally?

SME answer: A WHERE predicate on a nullable right-side column removes unmatched left rows.

Interviewer expects: *Justify the choice against alternatives and state tradeoffs.*

33. When should you use EXISTS?

SME answer: When testing whether a correlated match exists without returning joined duplicates.

Interviewer expects: *Specify assumptions, boundary conditions and validation checks.*

34. What is a correlated subquery?

SME answer: A nested query that references columns from an enclosing query block.

Interviewer expects: *Explain the behavior on realistic datasets and edge cases.*

35. How does LAG differ from LEAD?

SME answer: LAG reads a previous row; LEAD reads a following row in a defined window order.

Interviewer expects: *Discuss performance, concurrency or operational consequences when relevant.*

36. What is a recursive CTE?

SME answer: A CTE with an anchor and recursive member repeatedly deriving rows until termination.

Interviewer expects: *Describe a realistic failure mode and the steps to diagnose it.*

37. How do you query an employee hierarchy?

SME answer: Use a recursive CTE joining employee rows to parent employee identifiers.

Interviewer expects: *Justify the choice against alternatives and state tradeoffs.*

38. What is an execution plan?

SME answer: The DBMS's chosen access and join operations for executing a query.

Interviewer expects: *Specify assumptions, boundary conditions and validation checks.*

39. Why can SELECT * hurt performance?

SME answer: It reads unnecessary columns, increases transfer and can inhibit covering-index benefits.

Interviewer expects: *Explain the behavior on realistic datasets and edge cases.*

40. What is the difference between clustered and nonclustered indexes?

SME answer: Generally clustered storage orders data by index key; nonclustered indexes store separate lookup structures, DBMS-dependent.

Interviewer expects: *Discuss performance, concurrency or operational consequences when relevant.*

41. What is a transaction isolation level?

SME answer: A setting governing which concurrent changes a transaction can observe.

Interviewer expects: *Describe a realistic failure mode and the steps to diagnose it.*

42. What is a dirty read?

SME answer: Reading data written by another transaction before it commits.

Interviewer expects: *Justify the choice against alternatives and state tradeoffs.*

43. What is a deadlock?

SME answer: Two or more transactions wait cyclically for resources; the DBMS typically aborts one.

Interviewer expects: *Specify assumptions, boundary conditions and validation checks.*

44. What is an upsert?

SME answer: A conditional insert-or-update operation, with implementation and concurrency semantics varying by DBMS.

Interviewer expects: *Explain the behavior on realistic datasets and edge cases.*

45. How would you update inventory safely?

SME answer: Check stock and decrement within a transaction using guarded updates and proper locking.

Interviewer expects: *Discuss performance, concurrency or operational consequences when relevant.*

46. What makes a query non-sargable?

SME answer: Expressions on indexed columns can prevent efficient index seeks in some engines.

Interviewer expects: *Describe a realistic failure mode and the steps to diagnose it.*

47. What is a database backup strategy?

SME answer: A tested schedule of full/incremental or log backups aligned with recovery objectives.

Interviewer expects: *Justify the choice against alternatives and state tradeoffs.*

48. What is the difference between RPO and RTO?

SME answer: RPO is acceptable data loss; RTO is acceptable service restoration time.

Interviewer expects: *Specify assumptions, boundary conditions and validation checks.*

49. Why might OFFSET pagination be slow?

SME answer: Large offsets may require scanning or sorting skipped rows; keyset pagination may be faster.

Interviewer expects: *Explain the behavior on realistic datasets and edge cases.*

50. How do you benchmark a query optimization?

SME answer: Compare plans, elapsed time, reads and resource use under equivalent data and workloads.

Interviewer expects: *Discuss performance, concurrency or operational consequences when relevant.*

MODULE 03 | Data Warehousing, Snowflake and Data Modeling

What interviewers expect: correct terminology, a brief explanation, a practical illustration and awareness of common pitfalls.

MEDIUM | Concept clarity and core usage (Q1-Q25)

01. What is a data warehouse?

SME answer: A system optimized for integrated historical analytics and reporting rather than transactional writes.

Interviewer expects: *Explain the definition accurately and give a simple business example.*

02. What is OLTP versus OLAP?

SME answer: OLTP supports frequent short transactions; OLAP supports complex aggregations and analytical scans.

Interviewer expects: *Contrast the behavior with a related concept and identify the key limitation.*

03. What is a fact table?

SME answer: A table of measurable events or snapshots at a declared business grain.

Interviewer expects: *Describe a small working example and its expected output.*

04. What is a dimension table?

SME answer: A descriptive context table used to filter, group and label facts.

Interviewer expects: *State when the concept is useful and one common mistake.*

05. What is a star schema?

SME answer: A central fact table directly connected to denormalized dimension tables.

Interviewer expects: *Connect the concept to correctness, maintainability or performance.*

06. What is a snowflake schema?

SME answer: A dimensional model whose dimension attributes are normalized across related tables.

Interviewer expects: *Explain the definition accurately and give a simple business example.*

07. What is grain in dimensional modeling?

SME answer: The exact real-world event or entity represented by one fact row.

Interviewer expects: *Contrast the behavior with a related concept and identify the key limitation.*

08. What is a surrogate key?

SME answer: A warehouse-generated identifier independent of source business identifiers.

Interviewer expects: *Describe a small working example and its expected output.*

09. What is an SCD Type 1 change?

SME answer: Overwrite a dimension attribute without preserving its prior value.

Interviewer expects: *State when the concept is useful and one common mistake.*

10. What is an SCD Type 2 change?

SME answer: Insert a new versioned dimension row to preserve attribute history.

Interviewer expects: *Connect the concept to correctness, maintainability or performance.*

11. What is an SCD Type 3 change?

SME answer: Track limited previous values in extra columns alongside current values.

Interviewer expects: *Explain the definition accurately and give a simple business example.*

12. What is ETL versus ELT?

SME answer: ETL transforms before target loading; ELT loads first and transforms on the destination platform.

Interviewer expects: *Contrast the behavior with a related concept and identify the key limitation.*

13. What is a staging layer?

SME answer: A landing area for raw or lightly standardized source data before integration.

Interviewer expects: *Describe a small working example and its expected output.*

14. What is a data mart?

SME answer: A subject- or department-oriented subset of analytical data.

Interviewer expects: *State when the concept is useful and one common mistake.*

15. What is an enterprise data warehouse?

SME answer: A governed, integrated analytical repository spanning multiple business domains.

Interviewer expects: *Connect the concept to correctness, maintainability or performance.*

16. What is Snowflake?

SME answer: A managed cloud data platform with decoupled storage and virtual warehouse compute.

Interviewer expects: *Explain the definition accurately and give a simple business example.*

17. What is a Snowflake virtual warehouse?

SME answer: An independently sized compute resource used to run queries and data-loading work.

Interviewer expects: *Contrast the behavior with a related concept and identify the key limitation.*

18. What is a Snowflake database and schema?

SME answer: Logical containers organizing tables, views and other objects within an account.

Interviewer expects: *Describe a small working example and its expected output.*

19. What is a Snowflake stage?

SME answer: A named location or configuration used to access files for data loading or unloading.

Interviewer expects: *State when the concept is useful and one common mistake.*

20. What is COPY INTO in Snowflake?

SME answer: A command that bulk loads staged files into tables or unloads query output to a location.

Interviewer expects: *Connect the concept to correctness, maintainability or performance.*

21. Why use a landing-to-curated pipeline?

SME answer: It separates ingestion fidelity from business transformations and data quality controls.

Interviewer expects: *Explain the definition accurately and give a simple business example.*

22. What is batch loading?

SME answer: Processing data in bounded scheduled groups rather than as continuously arriving events.

Interviewer expects: *Contrast the behavior with a related concept and identify the key limitation.*

23. What is data profiling?

SME answer: Measuring distributions, null rates, uniqueness and validity before transformation.

Interviewer expects: *Describe a small working example and its expected output.*

24. What is data lineage?

SME answer: Recorded relationships connecting source data, transformations and downstream outputs.

Interviewer expects: *State when the concept is useful and one common mistake.*

25. Why reconcile source and target counts?

SME answer: To detect unintended omissions, duplicates and rejected rows after data movement.

Interviewer expects: *Connect the concept to correctness, maintainability or performance.*

ADVANCED | Scenarios, tradeoffs and troubleshooting (Q26-Q50)

26. How do you design an order-line fact table?

SME answer: Declare one row per order line, store measures and keys to customer, product and dates.

Interviewer expects: *Describe a realistic failure mode and the steps to diagnose it.*

27. How do you handle late-arriving dimensions?

SME answer: Use an unknown or inferred dimension member, then repair foreign keys when details arrive.

Interviewer expects: *Justify the choice against alternatives and state tradeoffs.*

28. How do you handle late-arriving facts?

SME answer: Load them against the correct event date and resolve historical dimension versions.

Interviewer expects: *Specify assumptions, boundary conditions and validation checks.*

29. Why are conformed dimensions valuable?

SME answer: Shared dimensions make cross-process metrics consistently comparable.

Interviewer expects: *Explain the behavior on realistic datasets and edge cases.*

30. What is a degenerate dimension?

SME answer: A transaction identifier stored in a fact without a separate descriptive dimension table.

Interviewer expects: *Discuss performance, concurrency or operational consequences when relevant.*

31. When should you use a snapshot fact table?

SME answer: For periodic states such as daily account balances or inventory levels.

Interviewer expects: *Describe a realistic failure mode and the steps to diagnose it.*

32. What is an accumulating snapshot?

SME answer: A fact row updated as milestones in a business process are reached.

Interviewer expects: *Justify the choice against alternatives and state tradeoffs.*

33. How do you avoid double counting after joins?

SME answer: Respect fact grain, use unique dimension keys and aggregate only after validating cardinality.

Interviewer expects: *Specify assumptions, boundary conditions and validation checks.*

34. How do you implement SCD Type 2 safely?

SME answer: Detect changes, close the current version and insert a new version in an atomic pattern.

Interviewer expects: *Explain the behavior on realistic datasets and edge cases.*

35. What is a current-row indicator in SCD2?

SME answer: A Boolean or status marking the active version of a business key.

Interviewer expects: *Discuss performance, concurrency or operational consequences when relevant.*

36. How can you test SCD2 validity?

SME answer: Check no overlapping effective-date ranges and exactly one active record per key.

Interviewer expects: *Describe a realistic failure mode and the steps to diagnose it.*

37. What is idempotent loading?

SME answer: Rerunning a load yields the same logical target state without duplicate effects.

Interviewer expects: *Justify the choice against alternatives and state tradeoffs.*

38. What is a watermark in incremental ETL?

SME answer: A persisted high-water value used to select newly changed source data.

Interviewer expects: *Specify assumptions, boundary conditions and validation checks.*

39. What risk does a timestamp watermark introduce?

SME answer: Late or out-of-order changes may be missed without overlap or change tracking.

Interviewer expects: *Explain the behavior on realistic datasets and edge cases.*

40. What is CDC?

SME answer: Change Data Capture records inserts, updates and deletes from operational sources.

Interviewer expects: *Discuss performance, concurrency or operational consequences when relevant.*

41. What are Snowflake micro-partitions?

SME answer: Automatically managed columnar storage units with metadata used for pruning.

Interviewer expects: *Describe a realistic failure mode and the steps to diagnose it.*

42. What does pruning mean in Snowflake?

SME answer: Skipping micro-partitions whose metadata proves they cannot satisfy query filters.

Interviewer expects: *Justify the choice against alternatives and state tradeoffs.*

43. When should you scale a warehouse up versus out?

SME answer: Scale up for per-query resources; multi-cluster scale out to address concurrent queries.

Interviewer expects: *Specify assumptions, boundary conditions and validation checks.*

44. What is Snowflake Time Travel?

SME answer: A retention-governed capability to query or recover historical table states.

Interviewer expects: *Explain the behavior on realistic datasets and edge cases.*

45. Why might Snowflake cloning be useful?

SME answer: Zero-copy cloning enables fast environment duplication with copy-on-write storage behavior.

Interviewer expects: *Discuss performance, concurrency or operational consequences when relevant.*

46. How should you protect sensitive warehouse data?

SME answer: Use least privilege, masking, row access controls and audited access policies.

Interviewer expects: *Describe a realistic failure mode and the steps to diagnose it.*

47. What is the difference between RBAC and object ownership?

SME answer: RBAC assigns privileges to roles; ownership controls certain administrative actions.

Interviewer expects: *Justify the choice against alternatives and state tradeoffs.*

48. How do you validate a finance data mart?

SME answer: Reconcile source totals, dimensions, grain, exception counts and period cutoffs.

Interviewer expects: *Specify assumptions, boundary conditions and validation checks.*

49. How do you evaluate warehouse query cost?

SME answer: Measure compute consumption, scan volume, query duration and workload concurrency.

Interviewer expects: *Explain the behavior on realistic datasets and edge cases.*

50. How do you build a production-ready batch pipeline?

SME answer: Include orchestration, retries, idempotence, validation, lineage, alerts and recovery steps.

Interviewer expects: *Discuss performance, concurrency or operational consequences when relevant.*

MODULE 04 | Python Programming and Software Design

What interviewers expect: correct terminology, a brief explanation, a practical illustration and awareness of common pitfalls.

MEDIUM | Concept clarity and core usage (Q1-Q25)

01. What is the difference between list and tuple?

SME answer: Lists are mutable sequences; tuples are immutable sequences.

Interviewer expects: *Explain the definition accurately and give a simple business example.*

02. What is a dictionary?

SME answer: A mapping from hashable keys to values, preserving insertion order in modern Python.

Interviewer expects: *Contrast the behavior with a related concept and identify the key limitation.*

03. What is a set?

SME answer: An unordered collection of distinct hashable elements.

Interviewer expects: *Describe a small working example and its expected output.*

04. What does == versus is mean?

SME answer: == tests value equality; is tests object identity.

Interviewer expects: *State when the concept is useful and one common mistake.*

05. What are mutable versus immutable objects?

SME answer: Mutable objects can change in place; immutable objects cannot change after creation.

Interviewer expects: *Connect the concept to correctness, maintainability or performance.*

06. What is a list comprehension?

SME answer: A concise expression producing a list from an iterable with optional filtering.

Interviewer expects: *Explain the definition accurately and give a simple business example.*

07. What is a generator?

SME answer: A lazy iterator producing values as requested, commonly using yield.

Interviewer expects: *Contrast the behavior with a related concept and identify the key limitation.*

08. What is *args versus **kwargs?

SME answer: *args gathers positional arguments; **kwargs gathers keyword arguments.

Interviewer expects: *Describe a small working example and its expected output.*

09. What is a lambda?

SME answer: An anonymous function restricted to a single expression.

Interviewer expects: *State when the concept is useful and one common mistake.*

10. What does enumerate do?

SME answer: It yields pairs of index and item while iterating.

Interviewer expects: *Connect the concept to correctness, maintainability or performance.*

11. What does zip do?

SME answer: It combines iterables into tuples of corresponding elements, stopping at the shortest by default.

Interviewer expects: *Explain the definition accurately and give a simple business example.*

12. What is exception handling?

SME answer: Handling anticipated failures with try, except, else and finally.

Interviewer expects: *Contrast the behavior with a related concept and identify the key limitation.*

13. What is a context manager?

SME answer: An object that manages entry/exit behavior, commonly used with with for safe cleanup.

Interviewer expects: *Describe a small working example and its expected output.*

14. What is a Python module?

SME answer: A file or module object containing reusable Python definitions.

Interviewer expects: *State when the concept is useful and one common mistake.*

15. What is a Python package?

SME answer: A namespace grouping modules; namespace packages may omit __init__.py.

Interviewer expects: *Connect the concept to correctness, maintainability or performance.*

16. What is a class?

SME answer: A blueprint/type defining shared behavior and state for its instances.

Interviewer expects: *Explain the definition accurately and give a simple business example.*

17. What is __init__?

SME answer: An initializer called after instance creation to configure initial state.

Interviewer expects: *Contrast the behavior with a related concept and identify the key limitation.*

18. What is inheritance?

SME answer: A class reuses or extends behavior from another class.

Interviewer expects: *Describe a small working example and its expected output.*

19. What is polymorphism?

SME answer: Different objects support a common interface with type-specific behavior.

Interviewer expects: *State when the concept is useful and one common mistake.*

20. What is encapsulation?

SME answer: Bundling behavior with state and controlling access via interfaces and conventions.

Interviewer expects: *Connect the concept to correctness, maintainability or performance.*

21. What is a property?

SME answer: An attribute interface backed by getter, setter or deletion methods.

Interviewer expects: *Explain the definition accurately and give a simple business example.*

22. What is the difference between append and extend?

SME answer: append adds one object; extend appends each element of an iterable.

Interviewer expects: *Contrast the behavior with a related concept and identify the key limitation.*

23. What is slicing?

SME answer: Selecting a portion of a sequence with start, stop and step indices.

Interviewer expects: *Describe a small working example and its expected output.*

24. What is duck typing?

SME answer: Compatibility determined by supported operations rather than declared nominal type.

Interviewer expects: *State when the concept is useful and one common mistake.*

25. What is an assertion?

SME answer: A developer check that raises AssertionError when false; not for production input validation.

Interviewer expects: *Connect the concept to correctness, maintainability or performance.*

ADVANCED | Scenarios, tradeoffs and troubleshooting (Q26-Q50)

26. Why is a mutable default argument dangerous?

SME answer: A default list or dict is created once and can leak state across calls.

Interviewer expects: *Describe a realistic failure mode and the steps to diagnose it.*

27. How do you fix a mutable default argument?

SME answer: Use None as the default and allocate a fresh mutable object inside the function.

Interviewer expects: *Justify the choice against alternatives and state tradeoffs.*

28. What is shallow copy versus deep copy?

SME answer: Shallow copying duplicates only the outer container; deep copying recursively duplicates nested content.

Interviewer expects: *Specify assumptions, boundary conditions and validation checks.*

29. How do you avoid catching every exception silently?

SME answer: Catch specific exception types and log or re-raise unexpected errors.

Interviewer expects: *Explain the behavior on realistic datasets and edge cases.*

30. What is the difference between raise and return?

SME answer: raise interrupts control flow with an exception; return finishes a function normally.

Interviewer expects: *Discuss performance, concurrency or operational consequences when relevant.*

31. What is the difference between iterable and iterator?

SME answer: An iterable can produce an iterator; an iterator maintains state and supports next().

Interviewer expects: *Describe a realistic failure mode and the steps to diagnose it.*

32. How do decorators work?

SME answer: A callable wraps another callable to add behavior without changing callers.

Interviewer expects: *Justify the choice against alternatives and state tradeoffs.*

33. What is recursion depth?

SME answer: The call-stack limit that bounds recursive execution in CPython unless configured differently.

Interviewer expects: *Specify assumptions, boundary conditions and validation checks.*

34. Why use type hints?

SME answer: They document interfaces and support static tooling without normally enforcing runtime types.

Interviewer expects: *Explain the behavior on realistic datasets and edge cases.*

35. What are dataclasses?

SME answer: A standard mechanism generating methods for classes primarily storing structured fields.

Interviewer expects: *Discuss performance, concurrency or operational consequences when relevant.*

36. What is MRO?

SME answer: Method Resolution Order defines how Python looks up methods in multiple inheritance.

Interviewer expects: *Describe a realistic failure mode and the steps to diagnose it.*

37. What is composition versus inheritance?

SME answer: Composition assembles collaborators; inheritance models substitutable is-a relationships.

Interviewer expects: *Justify the choice against alternatives and state tradeoffs.*

38. What is the GIL in CPython?

SME answer: A lock restricting simultaneous execution of Python bytecode in traditional GIL-enabled builds.

Interviewer expects: *Specify assumptions, boundary conditions and validation checks.*

39. When should you use multiprocessing?

SME answer: For CPU-bound parallelism that benefits from separate processes and can afford IPC overhead.

Interviewer expects: *Explain the behavior on realistic datasets and edge cases.*

40. How do you profile slow Python code?

SME answer: Measure representative workloads with cProfile and target high-cost call paths.

Interviewer expects: *Discuss performance, concurrency or operational consequences when relevant.*

41. What is a virtual environment?

SME answer: An isolated installation context for project-specific Python dependencies.

Interviewer expects: *Describe a realistic failure mode and the steps to diagnose it.*

42. Why pin dependencies?

SME answer: Pinned versions improve reproducibility, though security updates still require maintenance.

Interviewer expects: *Justify the choice against alternatives and state tradeoffs.*

43. What is pytest fixture usage?

SME answer: Fixtures supply reusable test setup and teardown through dependency injection.

Interviewer expects: *Specify assumptions, boundary conditions and validation checks.*

44. How do you mock a database in unit tests?

SME answer: Replace data access boundaries with controlled fakes or mocks and assert interactions.

Interviewer expects: *Explain the behavior on realistic datasets and edge cases.*

45. How do you prevent secrets leaking into scripts?

SME answer: Use environment-based secret stores, access controls and redacted logging.

Interviewer expects: *Discuss performance, concurrency or operational consequences when relevant.*

46. How do you handle file encoding issues?

SME answer: Specify appropriate text encodings and error handling instead of relying on locale defaults.

Interviewer expects: *Describe a realistic failure mode and the steps to diagnose it.*

47. What is structured logging?

SME answer: Emitting consistent machine-readable event fields for reliable analysis.

Interviewer expects: *Justify the choice against alternatives and state tradeoffs.*

48. How should retries be designed?

SME answer: Retry transient failures with bounded attempts, backoff and idempotent operations.

Interviewer expects: *Specify assumptions, boundary conditions and validation checks.*

49. How do you design testable ETL functions?

SME answer: Keep parsing and transformation logic pure and isolate I/O behind small interfaces.

Interviewer expects: *Explain the behavior on realistic datasets and edge cases.*

50. How can Python process large files safely?

SME answer: Stream records or chunks rather than reading entire files into memory.

Interviewer expects: *Discuss performance, concurrency or operational consequences when relevant.*

MODULE 05 | Python Analytics, Integration and Projects

What interviewers expect: correct terminology, a brief explanation, a practical illustration and awareness of common pitfalls.

MEDIUM | Concept clarity and core usage (Q1-Q25)

01. What is a pandas DataFrame?

SME answer: A two-dimensional labeled data structure supporting heterogeneous typed columns.

Interviewer expects: *Explain the definition accurately and give a simple business example.*

02. How do you read a CSV with pandas?

SME answer: Use pandas.read_csv with explicit types, parsing rules and null-value assumptions.

Interviewer expects: *Contrast the behavior with a related concept and identify the key limitation.*

03. What is Series versus DataFrame?

SME answer: A Series is one-dimensional labeled data; a DataFrame is tabular with multiple columns.

Interviewer expects: *Describe a small working example and its expected output.*

04. How do you handle missing values?

SME answer: Profile null meaning, then drop, impute, flag or retain based on the business rule.

Interviewer expects: *State when the concept is useful and one common mistake.*

05. What does fillna do?

SME answer: It replaces missing values with specified values or applicable filling logic.

Interviewer expects: *Connect the concept to correctness, maintainability or performance.*

06. How do you find duplicates in pandas?

SME answer: Use duplicated() or drop_duplicates() with a chosen key subset and retention rule.

Interviewer expects: *Explain the definition accurately and give a simple business example.*

07. What is groupby in pandas?

SME answer: A split-apply-combine operation that aggregates or transforms rows by group keys.

Interviewer expects: *Contrast the behavior with a related concept and identify the key limitation.*

08. What is a pandas merge?

SME answer: A relational-style join between DataFrames using defined keys and join types.

Interviewer expects: *Describe a small working example and its expected output.*

09. What is concat?

SME answer: An operation combining pandas objects along an axis with configurable alignment.

Interviewer expects: *State when the concept is useful and one common mistake.*

10. What is the difference between loc and iloc?

SME answer: loc selects by labels; iloc selects by integer positions.

Interviewer expects: *Connect the concept to correctness, maintainability or performance.*

11. What is NumPy ndarray?

SME answer: A homogeneous n-dimensional array enabling vectorized numerical operations.

Interviewer expects: *Explain the definition accurately and give a simple business example.*

12. What is broadcasting?

SME answer: NumPy's rules for applying elementwise operations across compatible shapes.

Interviewer expects: *Contrast the behavior with a related concept and identify the key limitation.*

13. What is vectorization?

SME answer: Applying array operations in optimized code instead of explicit Python element loops.

Interviewer expects: *Describe a small working example and its expected output.*

14. What is the difference between NumPy and pandas?

SME answer: NumPy emphasizes typed arrays; pandas adds labels and table-oriented operations.

Interviewer expects: *State when the concept is useful and one common mistake.*

15. What is a histogram?

SME answer: A chart showing the frequency distribution of numeric values across bins.

Interviewer expects: *Connect the concept to correctness, maintainability or performance.*

16. When would you use a scatter plot?

SME answer: To inspect relationships, clusters or outliers between two numeric variables.

Interviewer expects: *Explain the definition accurately and give a simple business example.*

17. What is matplotlib?

SME answer: A Python plotting library offering granular control over chart creation.

Interviewer expects: *Contrast the behavior with a related concept and identify the key limitation.*

18. What is seaborn?

SME answer: A statistical visualization library built on matplotlib with high-level plot interfaces.

Interviewer expects: *Describe a small working example and its expected output.*

19. How do you connect Python to SQLite?

SME answer: Use sqlite3.connect and parameterized SQL inside managed transactions.

Interviewer expects: *State when the concept is useful and one common mistake.*

20. What is SQLAlchemy?

SME answer: A Python SQL toolkit and ORM providing database connectivity and query abstractions.

Interviewer expects: *Connect the concept to correctness, maintainability or performance.*

21. What is JSON?

SME answer: A text interchange format supporting objects, arrays and basic scalar values.

Interviewer expects: *Explain the definition accurately and give a simple business example.*

22. How do you load JSON in Python?

SME answer: Use json.load for a file object or json.loads for a JSON string.

Interviewer expects: *Contrast the behavior with a related concept and identify the key limitation.*

23. What is XML parsing?

SME answer: Reading tree-structured tagged data with an appropriate parser and security settings.

Interviewer expects: *Describe a small working example and its expected output.*

24. How do you test a transformation?

SME answer: Assert schema, row counts, key constraints and known input-output examples.

Interviewer expects: *State when the concept is useful and one common mistake.*

25. Why use logging in a data pipeline?

SME answer: To capture run IDs, progress, data quality results and actionable errors.

Interviewer expects: *Connect the concept to correctness, maintainability or performance.*

ADVANCED | Scenarios, tradeoffs and troubleshooting (Q26-Q50)

26. How do you process a 10-GB CSV on a low-memory machine?

SME answer: Read in chunks or use a scalable engine; aggregate incrementally to bound memory.

Interviewer expects: *Describe a realistic failure mode and the steps to diagnose it.*

27. How do you avoid a cartesian join in pandas?

SME answer: Validate join keys and cardinality before merge, using validate where applicable.

Interviewer expects: *Justify the choice against alternatives and state tradeoffs.*

28. How do you speed up a slow pandas loop?

SME answer: Prefer vectorized operations, categorical types or group-based transformations.

Interviewer expects: *Specify assumptions, boundary conditions and validation checks.*

29. How do you detect numeric outliers?

SME answer: Use domain thresholds or methods such as IQR, while validating real anomalies.

Interviewer expects: *Explain the behavior on realistic datasets and edge cases.*

30. How do you calculate daily revenue?

SME answer: Parse event dates and aggregate valid monetary amounts by date and business grain.

Interviewer expects: *Discuss performance, concurrency or operational consequences when relevant.*

31. How do you avoid double-counting transactions?

SME answer: Deduplicate stable transaction identifiers before performing amount aggregations.

Interviewer expects: *Describe a realistic failure mode and the steps to diagnose it.*

32. What is a schema drift problem?

SME answer: Source fields or types change and break assumptions in downstream ingestion or transformations.

Interviewer expects: *Justify the choice against alternatives and state tradeoffs.*

33. How do you handle schema drift?

SME answer: Validate contracts, map compatible changes and quarantine incompatible records.

Interviewer expects: *Specify assumptions, boundary conditions and validation checks.*

34. How do you process a REST API reliably?

SME answer: Use pagination, timeouts, bounded retries, rate-limit handling and checkpointed progress.

Interviewer expects: *Explain the behavior on realistic datasets and edge cases.*

35. What is a data quality rule?

SME answer: An explicit test of correctness such as uniqueness, completeness, range or referential validity.

Interviewer expects: *Discuss performance, concurrency or operational consequences when relevant.*

36. How would you reconcile two data files?

SME answer: Compare schema, record counts, unique keys, aggregate totals and mismatched rows.

Interviewer expects: *Describe a realistic failure mode and the steps to diagnose it.*

37. What is parameterized SQL?

SME answer: Queries that bind data values separately from SQL text to prevent injection and quoting errors.

Interviewer expects: *Justify the choice against alternatives and state tradeoffs.*

38. What is the risk of pickle deserialization?

SME answer: Loading untrusted pickle content can execute malicious code.

Interviewer expects: *Specify assumptions, boundary conditions and validation checks.*

39. How do you manage large JSON arrays?

SME answer: Stream or paginate input, flatten only needed fields and avoid unnecessary copies.

Interviewer expects: *Explain the behavior on realistic datasets and edge cases.*

40. What is a rolling average?

SME answer: An aggregation over a moving ordered time window, typically using rolling().mean().

Interviewer expects: *Discuss performance, concurrency or operational consequences when relevant.*

41. How do you handle timezone-aware timestamps?

SME answer: Parse into timezone-aware datetimes, normalize for storage and convert for display.

Interviewer expects: *Describe a realistic failure mode and the steps to diagnose it.*

42. What is a unit test versus integration test?

SME answer: A unit test isolates logic; an integration test verifies collaborating components or systems.

Interviewer expects: *Justify the choice against alternatives and state tradeoffs.*

43. How do you structure an analytics project?

SME answer: Separate ingestion, validation, transformation, analysis and reporting into testable modules.

Interviewer expects: *Specify assumptions, boundary conditions and validation checks.*

44. What makes a visualization misleading?

SME answer: Truncated axes, wrong aggregation, inconsistent units or omitted uncertainty can distort meaning.

Interviewer expects: *Explain the behavior on realistic datasets and edge cases.*

45. How do you design incremental file processing?

SME answer: Persist processed-file or record offsets and ensure reruns do not duplicate output.

Interviewer expects: *Discuss performance, concurrency or operational consequences when relevant.*

46. How do you organize project configuration?

SME answer: Store environment-specific values separately from code and validate them at startup.

Interviewer expects: *Describe a realistic failure mode and the steps to diagnose it.*

47. What metrics would you monitor for a pipeline?

SME answer: Latency, throughput, freshness, failed records, retry counts and quality-rule pass rates.

Interviewer expects: *Justify the choice against alternatives and state tradeoffs.*

48. What should an error quarantine contain?

SME answer: Failed records, source metadata, reason codes, timestamps and traceable run identifiers.

Interviewer expects: *Specify assumptions, boundary conditions and validation checks.*

49. How do you demonstrate a Python analytics project in an interview?

SME answer: Explain the business question, dataset, cleaning logic, metrics, validation and measurable outcome.

Interviewer expects: *Explain the behavior on realistic datasets and edge cases.*

50. How do you make a data pipeline reproducible?

SME answer: Version code, dependencies, input contracts, transformations and execution configuration.

Interviewer expects: *Discuss performance, concurrency or operational consequences when relevant.*

MODULE 06 | Apache Spark and Distributed Processing

What interviewers expect: correct terminology, a brief explanation, a practical illustration and awareness of common pitfalls.

MEDIUM | Concept clarity and core usage (Q1-Q25)

01. What is Apache Spark?

SME answer: A distributed data processing engine supporting batch SQL, streaming and ML workloads.

Interviewer expects: *Explain the definition accurately and give a simple business example.*

02. What is an RDD?

SME answer: An immutable, partitioned and fault-tolerant collection processed across Spark workers.

Interviewer expects: *Contrast the behavior with a related concept and identify the key limitation.*

03. What is an RDD versus DataFrame?

SME answer: DataFrames expose schema-aware optimized execution; RDDs offer lower-level typed-object control.

Interviewer expects: *Describe a small working example and its expected output.*

04. What is Spark SQL?

SME answer: Spark's structured data engine accessible through SQL and DataFrame APIs.

Interviewer expects: *State when the concept is useful and one common mistake.*

05. What is lazy evaluation?

SME answer: Transformations build a plan and are evaluated when an action requires results.

Interviewer expects: *Connect the concept to correctness, maintainability or performance.*

06. What is a transformation?

SME answer: An operation producing a new logical dataset without immediately executing its full computation.

Interviewer expects: *Explain the definition accurately and give a simple business example.*

07. What is an action?

SME answer: An operation such as count or collect that requests execution and returns or writes results.

Interviewer expects: *Contrast the behavior with a related concept and identify the key limitation.*

08. What is Catalyst?

SME answer: Spark SQL's optimizer that rewrites and selects logical and physical plans.

Interviewer expects: *Describe a small working example and its expected output.*

09. What is Tungsten?

SME answer: A collection of Spark execution improvements including efficient memory handling and code generation.

Interviewer expects: *State when the concept is useful and one common mistake.*

10. What is a DAG?

SME answer: A directed acyclic graph representing dependencies among computations and stages.

Interviewer expects: *Connect the concept to correctness, maintainability or performance.*

11. What are narrow versus wide transformations?

SME answer: Narrow dependencies avoid all-to-all shuffle; wide dependencies require data redistribution.

Interviewer expects: *Explain the definition accurately and give a simple business example.*

12. What is SparkSession?

SME answer: The principal entry point for DataFrame, SQL and many Spark APIs.

Interviewer expects: *Contrast the behavior with a related concept and identify the key limitation.*

13. What is a broadcast join?

SME answer: A join strategy distributing a small relation to executors to avoid shuffling a large side.

Interviewer expects: *Describe a small working example and its expected output.*

14. What is checkpointing?

SME answer: Writing stable state to reliable storage to truncate lineage or support streaming recovery.

Interviewer expects: *State when the concept is useful and one common mistake.*

15. What is Structured Streaming?

SME answer: Spark's declarative streaming engine using DataFrame queries with incremental execution.

Interviewer expects: *Connect the concept to correctness, maintainability or performance.*

16. What is an accumulator?

SME answer: A distributed write-only-by-workers variable typically used for diagnostic counters.

Interviewer expects: *Explain the definition accurately and give a simple business example.*

17. What is partitioning?

SME answer: Distributing rows or records into work units across executors for parallel computation.

Interviewer expects: *Contrast the behavior with a related concept and identify the key limitation.*

18. How does Spark recover failed RDD partitions?

SME answer: It recomputes lost partitions using recorded lineage, when sources remain accessible.

Interviewer expects: *Describe a small working example and its expected output.*

19. What do cache and persist do?

SME answer: They retain reused datasets across actions in configured storage levels.

Interviewer expects: *State when the concept is useful and one common mistake.*

20. What is a Spark SQL window function?

SME answer: An analytic calculation over ordered or grouped rows without collapsing them.

Interviewer expects: *Connect the concept to correctness, maintainability or performance.*

21. What is an executor?

SME answer: A worker-side process running tasks and holding cached/shuffle data for an application.

Interviewer expects: *Explain the definition accurately and give a simple business example.*

22. What is a driver?

SME answer: The process coordinating Spark execution, constructing plans and scheduling work.

Interviewer expects: *Contrast the behavior with a related concept and identify the key limitation.*

23. What is a shuffle?

SME answer: Movement and repartitioning of records among executors to satisfy wide dependencies.

Interviewer expects: *Describe a small working example and its expected output.*

24. What is repartition versus coalesce?

SME answer: repartition reshuffles to a chosen partition count; coalesce often reduces partitions without full shuffle.

Interviewer expects: *State when the concept is useful and one common mistake.*

25. What is a job, stage and task?

SME answer: An action triggers a job, jobs contain shuffle-bounded stages, and each stage runs partition tasks.

Interviewer expects: *Connect the concept to correctness, maintainability or performance.*

ADVANCED | Scenarios, tradeoffs and troubleshooting (Q26-Q50)

26. How do you diagnose skewed joins?

SME answer: Inspect Spark UI task-duration disparities and large skewed partition sizes.

Interviewer expects: *Describe a realistic failure mode and the steps to diagnose it.*

27. How do you mitigate data skew?

SME answer: Use AQE skew-join handling, salting or better partition distribution as appropriate.

Interviewer expects: *Justify the choice against alternatives and state tradeoffs.*

28. What is AQE?

SME answer: Adaptive Query Execution adjusts parts of query plans using runtime statistics.

Interviewer expects: *Specify assumptions, boundary conditions and validation checks.*

29. What is predicate pushdown?

SME answer: Pushing applicable filters closer to data sources to reduce data read and processed.

Interviewer expects: *Explain the behavior on realistic datasets and edge cases.*

30. What is partition pruning?

SME answer: Skipping entire source partitions excluded by filter predicates.

Interviewer expects: *Discuss performance, concurrency or operational consequences when relevant.*

31. What are small-file problems?

SME answer: Excessive tiny files increase metadata overhead and task scheduling costs.

Interviewer expects: *Describe a realistic failure mode and the steps to diagnose it.*

32. How do you optimize small files?

SME answer: Compact files appropriately and tune write partitioning to control output file sizes.

Interviewer expects: *Justify the choice against alternatives and state tradeoffs.*

33. What is the risk of collect()?

SME answer: It transfers all result rows to the driver and can exhaust driver memory.

Interviewer expects: *Specify assumptions, boundary conditions and validation checks.*

34. What is explain() used for?

SME answer: It inspects logical and physical plans to diagnose scans, exchanges and joins.

Interviewer expects: *Explain the behavior on realistic datasets and edge cases.*

35. What is the difference between cache and checkpoint?

SME answer: Caching speeds reuse but retains lineage; checkpointing stores durable state and cuts lineage.

Interviewer expects: *Discuss performance, concurrency or operational consequences when relevant.*

36. How do you choose a broadcast threshold?

SME answer: Use actual small-side size, memory budget and Spark adaptive settings rather than blindly broadcasting.

Interviewer expects: *Describe a realistic failure mode and the steps to diagnose it.*

37. What is watermarked event-time processing?

SME answer: A lateness bound that helps stateful streaming queries manage delayed events and state cleanup.

Interviewer expects: *Justify the choice against alternatives and state tradeoffs.*

38. What is a streaming checkpoint directory?

SME answer: Persistent metadata and state used to recover progress and some stateful query operations.

Interviewer expects: *Specify assumptions, boundary conditions and validation checks.*

39. What is exactly-once processing in Spark?

SME answer: End-to-end exactly-once outcomes depend on source, sink, checkpointing and idempotent/transactional behavior.

Interviewer expects: *Explain the behavior on realistic datasets and edge cases.*

40. What is foreachBatch?

SME answer: A Structured Streaming sink hook that processes each micro-batch as a DataFrame.

Interviewer expects: *Discuss performance, concurrency or operational consequences when relevant.*

41. What is the difference between batch and micro-batch?

SME answer: Batch processes bounded inputs; micro-batch processes successive small bounded increments.

Interviewer expects: *Describe a realistic failure mode and the steps to diagnose it.*

42. Why can groupByKey be expensive?

SME answer: It may move entire groups across the network; pre-aggregation can reduce shuffle volume.

Interviewer expects: *Justify the choice against alternatives and state tradeoffs.*

43. What is a UDF performance tradeoff?

SME answer: Custom functions can limit optimizer visibility or add serialization overhead versus built-in expressions.

Interviewer expects: *Specify assumptions, boundary conditions and validation checks.*

44. How do you deduplicate incoming streaming records?

SME answer: Use event IDs and appropriate watermark/state handling, or idempotent sink upserts.

Interviewer expects: *Explain the behavior on realistic datasets and edge cases.*

45. What is a spill in Spark?

SME answer: Intermediate data exceeding execution memory is written to disk, increasing I/O.

Interviewer expects: *Discuss performance, concurrency or operational consequences when relevant.*

46. What causes executor out-of-memory errors?

SME answer: Oversized partitions, skew, caching and memory-heavy joins can exceed executor capacity.

Interviewer expects: *Describe a realistic failure mode and the steps to diagnose it.*

47. How do you tune shuffle partitions?

SME answer: Start with observed partition sizes and task counts; use AQE where available.

Interviewer expects: *Justify the choice against alternatives and state tradeoffs.*

48. How does Parquet help Spark?

SME answer: Its columnar format supports selective column reads, compression and predicate pushdown.

Interviewer expects: *Specify assumptions, boundary conditions and validation checks.*

49. How do you validate a Spark ingestion job?

SME answer: Check schema, counts, duplicates, null thresholds and reconciled aggregates.

Interviewer expects: *Explain the behavior on realistic datasets and edge cases.*

50. How would you troubleshoot a slow Spark job?

SME answer: Inspect Spark UI, physical plan, shuffle volume, skew, input size and resource utilization.

Interviewer expects: *Discuss performance, concurrency or operational consequences when relevant.*

FRESHER INTERVIEW EVALUATION GUIDE

Concept correctness (40%): Correct terminology and logical explanation without memorized inaccuracies.

Practical application (30%): An appropriate example, simple SQL/Python/pseudocode technique or business case.

Reasoning and tradeoffs (20%): Identification of assumptions, NULLs/duplicates, edge cases and performance implications.

Communication (10%): Structured, concise answers; transparent acknowledgment of uncertainty.

Suggested interview drill: 10 rapid concept questions (1 minute each), two scenario questions (4 minutes each), and one brief live exercise (10 minutes). Score with the criteria above.